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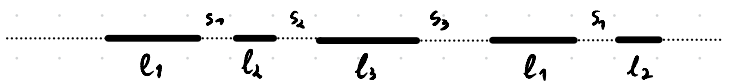
One-Dimensional Capacitance Matrix

In today's lecture, we will develop the theory for the one-dimensional capacitance matrix. Given a string of resonators with lengths l_i and spacings s_i



the resonance problem is as follows: find (ω, u) s.t.

$$\begin{cases} \left(\frac{d^2}{dx^2} + \omega^2 \gamma^{-2}\right) u = 0 & \text{in } D \\ \left(\frac{d^2}{dx^2} + \omega^2 \gamma_b^{-2}\right) u = 0 & \text{in } \mathbb{R} \setminus \bar{D} \\ u|_- = u|_+ & \text{in } \partial D \\ \delta u'|_- = u|_+ & \text{in } \partial D \\ \lim \left(\frac{d}{d|x|} - i\omega\right) u = 0 \end{cases}$$

$u|_\pm = \lim_{s>0,\, s\to 0} u(x \pm s)$

To treat this problem, we will not use layer potential techniques. Instead, there is a very explicit approach using the Dirichlet-to-Neumann operator, which is defined as follows

$$\begin{aligned} T^\omega : H^{1/2}(\partial D) &\longrightarrow H^{-1/2}(\partial D) \\ f &\longrightarrow \partial_n u_f \end{aligned}$$

where u_f is the solution to

$$\begin{cases} \left(\frac{d^2}{dx^2} + \omega^2\right) u_f = 0 & \text{in } D \\ u = f & \text{in } \partial D \\ \lim \left(\frac{d}{d|x|} - i\omega\right) u = 0 \end{cases}$$

That is, T^ω maps the data for the Dirichlet problem to that of the Neumann problem.
If we denote the endpoints of the ith resonator as (x_i^-, x_i^+), one can show that for $\omega \neq 0$ the DtN operator has the following form:

$$\begin{pmatrix} \gamma^w [f]_1^- \\ \gamma^w [f]_1^+ \\ \vdots \\ \gamma^w [f]_n^- \\ \gamma^w [f]_n^+ \end{pmatrix} = \begin{pmatrix} iw & & & \\ & A^w(s_1) & & \\ & & \ddots & \\ & & & A^w(s_{n-1}) \\ & & & iw \end{pmatrix} \begin{pmatrix} f_1^- \\ f_1^+ \\ \vdots \\ f_n^- \\ f_n^+ \end{pmatrix}$$

where

$$A^w(l) = \frac{1}{\sin(ws)} \begin{pmatrix} w\cos(sw) & w \\ w & -\cos(ws) \end{pmatrix}$$

To see this, lets consider the ith gap

$$\underset{x_i^+}{\overset{l_i}{\overline{\qquad\qquad}}} \; s_i \; \underset{x_{i+1}^-}{\overset{l_{i+1}}{\overline{\qquad\qquad}}}$$

Then the solution to

$$\begin{cases} \left(\frac{d^2}{dx^2} + w^2\right) u = 0 \\ u(x_i^+) = f_i^+ \\ u(x_{i+1}^-) = f_{i+1}^- \end{cases}$$

Can be written as a superposition of $e^{\pm iwx}$.

$$u = A e^{iwx} + B e^{-iwx}$$

This leads to

$$\begin{pmatrix} e^{iwx_i^+} & e^{-iwx_i^+} \\ e^{iwx_{i+1}^-} & e^{-iwx_{i+1}^-} \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} f_i^+ \\ f_{i+1}^- \end{pmatrix}$$

The inverse of this matrix is

$$\frac{-1}{2i\sin(\omega s)} \begin{pmatrix} e^{-i\omega x_{i+1}^-} & -e^{-i\omega x_i^+} \\ -e^{i\omega x_{i+1}^-} & e^{i\omega x_i^+} \end{pmatrix}$$

The derivative of u is

$$u'(x) = i\omega (A e^{i\omega x} - B e^{-i\omega x})$$

Since we need the normal derivatives, this can be written as

$$\begin{pmatrix} u'(x_i^+) \\ -u'(x_{i+1}^-) \end{pmatrix} = i\omega \begin{pmatrix} e^{i\omega x_i^+} & -e^{-i\omega x_i^+} \\ -e^{i\omega x_{i+1}^-} & e^{-i\omega x_{i+1}^-} \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix}$$

$$= A^\omega(s_i) \begin{pmatrix} f_i^+ \\ f_{i+1}^- \end{pmatrix}$$

At the left boundary, the solution must be a multiple of $e^{i\omega x}$ due to the radiation condition.

Take $\varphi \in C_c^\infty(\mathbb{R})$ and test the resonance problem

$$0 = -\int_D \left(\frac{d}{dx^2} + \omega^2 \gamma^2\right) u \bar{\varphi} \, dx$$

$$= \int_D u' \bar{\varphi}' - \omega^2 \gamma^2 u \bar{\varphi} \, dx + \sum_i u'|_+(x_i^+) \bar{\varphi}(x_i^+) - u'|_-(x_i^-) \bar{\varphi}(x_i^-)$$

$$= \int_D u' \bar{\varphi}' - \omega^2 \gamma^2 u \bar{\varphi} \, dx + \delta \sum_i u'|_-(x_i^+) \bar{\varphi}(x_i^+) - u'|_+(x_i^+) \bar{\varphi}(x_i^-)$$

$$= \int_D u' \bar{\varphi}' - \omega^2 \gamma^2 u \bar{\varphi} \, dx + \delta \langle \hat{T}^{\omega/v_b}[u], \varphi \rangle_{\partial D}$$

Taking the limit $\delta \to 0$, we find that

$$-\omega_0^2 a\varphi$$

$$\int_0 u'\varphi' \, dx = 0 \quad \forall \varphi \in C_c^\infty(\mathbb{R}) \longrightarrow u' = 0$$

where we used that we are looking for subwavelength resonances satisfying $\omega^2 \sim \delta$. Write

$$e_i = \begin{cases} 1 & x \in D_i \\ 0 & \text{else} \end{cases}$$

Since we can extend e_i to a function in $C_c^\infty(\mathbb{R})$, it must hold that

$$-\omega^2 \gamma^{-2} \int_{D_i} u \, dx + \delta \langle T^{\omega/\gamma_0}[u], e_i \rangle = 0$$

In particular, the same must be true for any linear combinations of e_i. Dividing by δ, we thus obtain the limiting problem

$$(C - \lambda V) u = 0$$

where

$$C_{ij} = \langle T^0[e_i], e_j \rangle, \quad V = \operatorname{diag}(\ell_i) \text{ and } \lambda = \omega^2/\gamma^{-2}$$

This reduction is of course a bit hand-wavy, but it can be made rigorous - perhaps we will do this in the beginning of December.

Taking the limit $\omega \to 0$ of T^ω, we obtain

$$\Upsilon^\omega = \begin{pmatrix} 0 & & & \\ & A^0(s_1) & & \\ & & \ddots & \\ & & & A^0(s_{n-1}) \\ & & & & 0 \end{pmatrix} + O(\omega^2)$$

where

$$A^0(s) = \begin{pmatrix} -1/s & 1/s \\ 1/s & -1/s \end{pmatrix}$$

Hence

$$C = \begin{pmatrix} \frac{1}{s_1} & -\frac{1}{s_1} & & & \\ -\frac{1}{s_1} & \frac{s_1+s_2}{s_1 s_2} & \ddots & & \\ & \ddots & \ddots & -\frac{1}{s_{n-2}} & \\ & & -\frac{1}{s_{n-2}} & \frac{1}{s_{n-2}} + \frac{1}{s_{n-1}} \end{pmatrix}$$

is symmetric and tridiagonal. Moreover,

$$V^{-1} C \sim \tilde{C} := V^{-1/2} C V^{1/2}$$

Hence we are interested in the eigenvalues of $\tilde{C}$, which is again symmetric and tridiagonal.

Laurent & Toeplitz Operators: Let $f \in L^\infty(S^1, \mathbb{C}^k)$, we then define the operator

$$M_f : L^2(S^1, \mathbb{C}^k) \longrightarrow L^2(S^1, \mathbb{C}^k)$$
$$u(x) \longrightarrow f(x)\, u(x)$$

This is a bounded linear operator on $L^2(S^1, \mathbb{C}^k)$

Write now f, u in terms of their Fourier coefficients

$$f(z) = \sum_{k \in \mathbb{Z}} A_k z^k, \qquad u(z) = \sum_{k \in \mathbb{Z}} u_k z^k$$

Then

$$M_f u(z) = \sum_{k \in \mathbb{Z}} z^k \sum_{\ell \in \mathbb{Z}} A_{k-\ell} u_\ell$$

That is

$$\begin{pmatrix} \vdots \\ (M_f u)_0 \\ \vdots \end{pmatrix} = \underbrace{\begin{pmatrix} \ddots & \ddots & \ddots & \ddots \\ \ddots & A_0 & A_{-1} & A_{-2} & \ddots \\ \ddots & A_1 & A_0 & A_{-1} & \ddots \\ \ddots & A_2 & A_1 & A_0 & \ddots \\ & \ddots & \ddots & \ddots & \ddots \end{pmatrix}}_{L(f)} \begin{pmatrix} \vdots \\ u_{-1} \\ u_0 \\ u_1 \\ \vdots \end{pmatrix}$$

Fourier transformation thus turns M_f into the Laurent operator $L(f)$. We call f the symbol of $L(f)$.

Denote now $P : \ell^2(\mathbb{Z}, \mathbb{C}^k) \longrightarrow \ell^2(\mathbb{N}, \mathbb{C}^k)$ the natural projection and $P^* : \ell^2(\mathbb{N}, \mathbb{C}^k) \longrightarrow \ell^2(\mathbb{Z}, \mathbb{C}^k)$ its adjoint. Consider

$$T(f) := P L(f) P^* = \begin{pmatrix} A_0 & A_{-1} & A_{-2} & \ddots \\ A_1 & A_0 & A_{-1} & \ddots \\ \ddots & \ddots & \ddots & \ddots \end{pmatrix}$$

This is the Toeplitz operator associated with f. Physically, one can think of $T(f)$ as describing the interactions of a semi-infinite system.

Finally, defining $P_n : \ell^2(\mathbb{Z}, \mathbb{C}^k) \longrightarrow \ell^2(\{1,\dots,n\}, \mathbb{C}^k)$ the natural projection, one defines the Toeplitz matrices as

$$T_n(f) = P_n \mathcal{L}(f) P_n^* = \begin{pmatrix} A_0 & A_1 & & \\ A_{-1} & A_0 & \ddots & \\ & \ddots & & A_1 \\ & & A_{-1} & A_0 \end{pmatrix}$$

To properly justify studying the infinite systems, we have to show that finite but large systems behave similar to them. Since our finite capacitance matrices are nothing but $T_n(f)$ with perturbations in the top left and bottom right corners (radiation cond.) understanding the behaviour of $T_n(f)$ as $n \to \infty$ already goes a long way.

As a first question, we may work how $\sigma(\mathcal{L}(f))$ and $\sigma(T(f))$ relate to one another. Using the duality with M_f (that is Floquet-Bloch!) we find that

$$\sigma(\mathcal{L}(f)) = \{\lambda \in \mathbb{C} \mid \exists z \in S^1 : \det(f(z) - \lambda) = 0\}$$

Our approximate eigenfunctions argument from last lecture furthermore yields

$$\sigma(T(f)) \supset \sigma(\mathcal{L}(f))$$

What about the converse inclusion? Consider $f(z) = z^{-1}$, then $\sigma(\mathcal{L}(f)) = S^1$ and

$$T(f) = \begin{pmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & \ddots & \ddots \end{pmatrix}$$

Let $u_n = 2^{1-n}$ and observe that

$$(T(f)-1/2)\,u = \begin{pmatrix} -1/2 & 1 & & \\ & -1/2 & 1 & \\ & & \ddots & \ddots \end{pmatrix} \begin{pmatrix} 1 \\ 1/2 \\ 1/4 \\ \vdots \end{pmatrix} = 0$$

hence $1/2 \in \sigma(T(f))$ and thus $\sigma(T(f)) \not\supset \sigma(L(f))$.
What went wrong? Well $u_n = 2^{-n}$ is formally also a solution to

$$(L(f) - 1/2)\,u_n = 0$$

However, u_n grows exponentially as $n \to -\infty$, hence it cannot be used to construct approximate eigenfunctions.
There is a more general underlying phenomenon. Write

$$f(z) = A_{-1} z^{-1} + A_0 + A_1 z$$

(This is always possible for bounded symbols!) and suppose $z, \lambda \in \mathbb{C}$, $v \in \mathbb{C}^4$ s.t.

$$f(z)\,v = \lambda v$$

Then, formally $v_n := z^n v$ satisfies

$$L(f)\,v = \begin{pmatrix} z^{-1}(A_{-1} z^{-1} + A_0 + A_1 z)\,v \\ (A_{-1} z^{-1} + A_0 + A_1 z)\,v \\ z(A_{-1} z^{-1} + A_0 + A_1 z)\,v \\ \vdots \end{pmatrix} = \lambda\, v$$

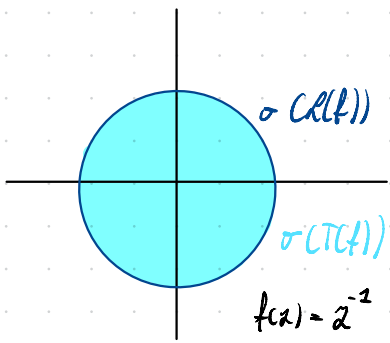
If $|z| < 1$, this vector decays as $n \to \infty$. Hence, if $\det f(z) - \lambda = 0$ has $> r$ solutions in $B_1(0)$, we can find an eigenfunction of $T(f)$ that decays exponentially.
Noting that $\det f(z) - \lambda$ generally has a pole of order r at 0, we find that $\lambda \notin \sigma(L(f))$

is an eigenvalue of $T(f)$ if

$$\mathrm{wind}(\det(f(z)-\lambda),0) > 0$$

while if $\mathrm{wind}(\det(f(z)-\lambda),0) < 0$, it is an eigenvalue of $T(f)^*$. Hence

$$\sigma(T(f)) \supset \sigma(\mathcal{L}(f)) \cup \{\lambda \in \mathbb{C} \setminus \mathcal{L}(f) \mid \mathrm{wind}(\det f(z)-\lambda,0) \neq 0\}$$



If $f(z) = \sum_{n=-p}^{q} a_n z^n$, then the following holds

Coburn's lemma: Suppose $\lambda \notin \sigma(\mathcal{L}(f))$ and f as above. Then it holds that

$$T(f)-\lambda \text{ is invertible} \iff \mathrm{wind}(f,\lambda) = 0.$$

And thus the description of the spectrum is an equality. This is in general not true for matrix valued symbols!

Consider

$$f(z) = \begin{pmatrix} 0 & 1+2z \\ 1+2z^{-2} & 1 \end{pmatrix} \quad \text{with} \quad \det f(z) = -(1+2z)(1+2z^{-2})$$

hence $\mathrm{wind}(\det f, 0) = 0$. But writing $u = (1, 0, -1/2, 0, 1/4, \dots)$ we have

$$T(f)\,u = \begin{pmatrix} 0 & 1 & & & \\ 1 & 1 & 2 & & \\ & 2 & 0 & 1 & \\ & & 1 & 1 & \\ & & & & \ddots \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ -1/2 \\ 0 \\ \vdots \end{pmatrix} = 0$$

Having understood the difference between $\sigma(R(f))$ and $\sigma(T(f))$, we now turn to

$$\sigma_{ess}(f) := \lim_{n\to\infty} \sigma(T_n(f))$$

Strong convergence immediately yields

$$\sigma_{ess}(f) \supset \sigma(T(f))$$

Once again, this is not an equality. The previous example is illustrative: If we let

$$f(z) = \begin{pmatrix} 1 & 1+2z \\ 1+2z^{-2} & 0 \end{pmatrix} \quad \text{with} \quad \det f(z) = -(1+2z)(1+2z^{-2})$$

Then $u = (1, 0, -2, 0, \dots)$ is formally an eigenvector of $T(f)$, but grows exponentially. However, for finite truncations of $T(f)$ we have

$$T_n(f)P_n u = \begin{pmatrix} 1 & 1 & & & & \\ 1 & 0 & 2 & & & \\ & 2 & \ddots & \ddots & & \\ & & \ddots & 2 & & \\ & & & 2 & 1 & 1 \\ & & & & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 2^{-n+1} \\ \vdots \\ -1/2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2^{-n+2} \\ 0 \\ \vdots \\ \vdots \end{pmatrix}$$

Since $T_n(f)$ is self adjoint, $\operatorname{dist}(\sigma(T_n(f)), 0) \to 0$, but $0 \notin T(f)$, since the formal eigenvector is

$$u = (1, -1, -1/2, 2, \dots)$$

which grows exponentially. Physically, it again makes sense: a finite system sees two boundaries, whereas $T(f)$ only sees one. In general, describing $\sigma_{ess}(f)$ thus remains a challenging task.

Sources & Further Reading

i) Subwavelength Resonances in One-Dimensional High-Contrast Acoustic Media, Florian Feppon, Zijian Cheng, Habib Ammari, 2023

ii) Introduction to Large Truncated Toeplitz Matrices, Albrecht Böttcher, Bernd Silbermann